What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?
ANS:

Factors to Consider When Selecting CPU, RAM, and Storage

Component	Factors to Consider
CPU	Processor speed, number of cores, motherboard compatibility, socket type, and power consumption.
RAM	Capacity (8 GB, 16 GB, etc.), DDR generation, supported speed, motherboard compatibility, and available slots.
Storage	Capacity, SSD or HDD type, NVMe/SATA interface, read/write speed, and motherboard compatibility.

Other important factors:

1. Purpose: Select components based on gaming, office work, programming, or video editing.
2. Budget: Choose components that provide good performance within your budget.
3. Upgrade options: Ensure the system supports future upgrades.
4. Cooling and power: Check CPU cooling requirements and power supply capacity.

Conclusion: Choosing compatible CPU, RAM, and storage ensures better performance, reliability, and future upgrade flexibility.

Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Antivirus software can slow down a computer in three ways:

1. CPU: Scanning files and detecting viruses uses processor resources, making applications run more slowly.

2. RAM: Antivirus software consumes memory while running in the background, leaving less RAM available for other applications.
3. Storage: Scanning files increases disk activity, while antivirus updates and quarantine files occupy storage space.

Conclusion: Antivirus software may cause temporary system slowdowns, especially during full scans or on computers with limited resources. Regular updates and scheduled scans can help maintain good performance.

Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues
ANS:

Troubleshooting Driver Incompatibility and Operating System Issues

1. Identify the problem: Check error messages, system crashes, or devices that are not working properly.
2. Check drivers: Open Device Manager and look for devices with warning symbols.
3. Update drivers: Download compatible drivers from the manufacturer's official website and install them.
4. Roll back drivers: If the problem started after a driver update, use the Roll Back Driver option in Device Manager.
5. Update the operating system: Install the latest compatible OS updates and security patches.

Run system checks: On Windows, open Command Prompt as Administrator and run:
sfc /scannow

6. DISM /Online /Cleanup-Image /RestoreHealth
7. Restart and test: Restart the computer and check whether the problem is resolved.

Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.
ANS:

How UPS and Surge Protectors Protect Computers

1. UPS (Uninterruptible Power Supply)

- Provides temporary battery power during a power failure.
- Allows users to save their work and shut down the computer safely.
- Prevents sudden shutdowns that can cause data loss or file corruption.

2. Surge Protector

- Protects computer components from sudden voltage spikes.
- Helps prevent damage to the motherboard, CPU, power supply, and other hardware.
- Reduces the risk of hardware failure caused by electrical surges.

Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.
ANS:

DNS and DHCP work together to provide network connectivity:

- DHCP (Dynamic Host Configuration Protocol): Automatically assigns IP addresses, subnet masks, default gateways, and DNS server addresses to devices.
- DNS (Domain Name System): Converts domain names, such as `www.google.com`, into IP addresses so devices can access websites.

Most likely misconfigured service: If devices can access websites using IP addresses but not domain names, the DNS service is most likely misconfigured or unavailable.

Solution: Verify the DNS server address provided by DHCP, check DNS records and server availability, and test resolution using `nslookup`.

Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.
ANS:

roubleshooting Firewall Rules Blocking Legitimate Traffic

1. Identify the issue: Check which applications, websites, or network services cannot connect.

2. Check firewall rules: Review inbound and outbound rules to identify blocked ports, IP addresses, or protocols.

3. Check firewall logs: Look for denied connections and confirm whether legitimate traffic is being blocked.

4. Test connectivity: Use commands such as `ping`, `tracert`, or `Test-NetConnection` to identify connection problems.

5. Modify firewall rules: Create or correct rules to allow legitimate traffic on the required ports and protocols.

6. Retest and monitor: Verify that the application works correctly and monitor logs for further issues.

Conclusion: Correctly configured firewall rules allow legitimate traffic while blocking unauthorized access. Avoid disabling the firewall completely; make only the necessary changes.

Task 1: Workstation Assembly Physically assemble a workstation for the startup office and document the component installation sequence. Install CPU, RAM, storage, and other core components into the chassis in the correct sequence. Connect all required internal cables (power, data, front panel). Document each installation step in order. System must pass POST (Power-On Self-Test) after assembly
ANS:

Component installation sequence

1. Prepare the chassis: Switch off the power supply, unplug it, and use an anti-static wrist strap.
2. Install the CPU: Fit the compatible CPU into the motherboard socket and lock it.
3. Install the CPU cooler: Apply thermal paste if required and secure the cooler. Connect its fan to `CPU_FAN`.
4. Install RAM: Insert compatible RAM modules into the recommended motherboard slots.
5. Install storage: Mount the SSD/HDD and connect SATA data and power cables, or install an M.2 SSD.
6. Install the motherboard: Fit it into the chassis using the correct standoffs and screws.
7. Install the power supply: Secure the PSU and connect the 24-pin motherboard and 8-pin CPU power cables.
8. Connect front-panel cables: Connect the power switch, reset switch, power LED, and USB/audio headers.
9. Install other components: Fit the graphics card or other expansion cards if required.
10. Final inspection: Check all connections, ensure no loose screws remain, and close the chassis.

POST verification

1. Connect the monitor, keyboard, and power cable.
2. Switch on the workstation.
3. Check that the system starts and displays the BIOS/UEFI screen without POST errors.
4. Verify that the CPU, RAM, and storage are detected correctly.

Task 2: RAM Mismatch Troubleshooting Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings on a startup office workstation. Identify the RAM mismatch error via BIOS/UEFI POST messages. Determine the root cause (mismatched speed, timing, or capacity). Resolve the mismatch by reconfiguring or reseating the RAM correctly. Confirm the system boots normally after the fix.
ANS:

Task 2: RAM Mismatch Troubleshooting

1. Identify the error

- Start the workstation and enter BIOS/UEFI by pressing `Del` or `F2`.
- Check POST messages for RAM errors, memory detection failures, or memory speed warnings.

2. Determine the root cause

- Check whether RAM modules have different speeds, timings, or capacities.
- Verify that the RAM type (DDR4/DDR5) is supported by the motherboard.
- Check the motherboard manual for compatible RAM configurations.

3. Resolve the mismatch

- Power off and unplug the workstation.
- Reseat the RAM modules firmly in the recommended slots.
- If modules are incompatible, use supported, matching RAM modules.
- In BIOS/UEFI, disable XMP/EXPO or load default memory settings to use compatible speeds.

4. Verify the fix

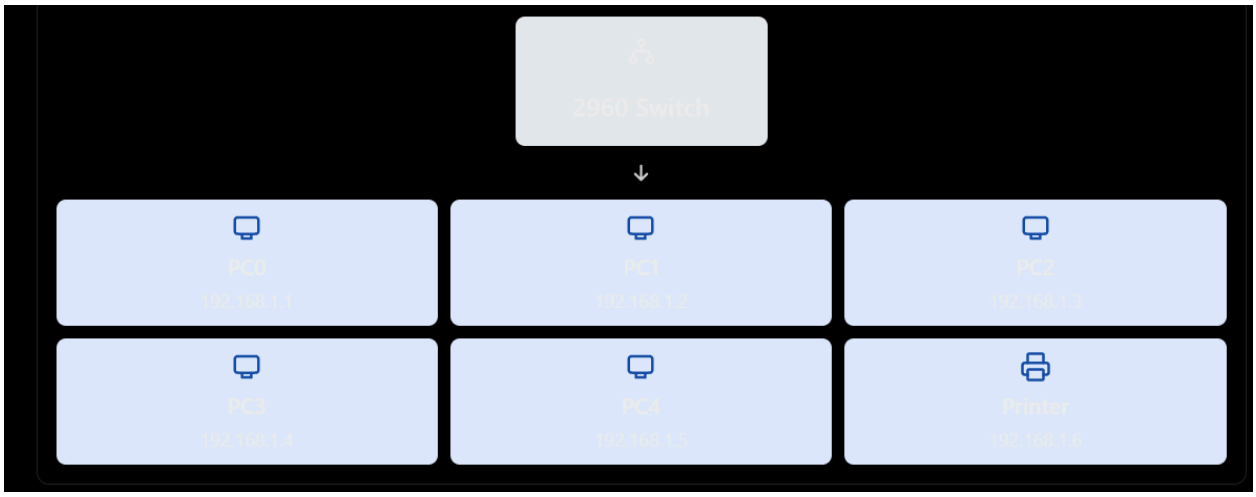
- Restart the workstation and check that POST completes without RAM errors.
- Confirm the correct memory capacity is detected in BIOS.
- Boot into the operating system and run a memory diagnostic if needed.

Task 3: LAN Topology Build Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet. Use Cisco Packet Tracer to build the topology. Include 5 end-systems and 1 network printer. Assign IP addresses from a Class C subnet to all devices. All assigned IPs must be reachable via Ping.
ANS:

Task 3: LAN Topology Build — Cisco Packet Tracer

1. Build the topology

Add 1 switch, 5 PCs, and 1 printer in Cisco Packet Tracer. Connect all devices to the switch using Copper Straight-Through cables.



2. Assign Class C IP addresses

Configure each PC via Desktop → IP Configuration → Static. Configure the printer through its IP configuration settings.

Device	IP address	Subnet mask

PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
Printer	192.168.1.6	255.255.255.0

Default gateway is not required for communication within this LAN.

3. Test connectivity

On PC0, open Desktop → Command Prompt and run:

```
ping 192.168.1.2
ping 192.168.1.3
ping 192.168.1.4
ping 192.168.1.5
ping 192.168.1.6
```

Task 4: Clean OS Installation Perform a clean OS installation with manual disk partitioning and startup optimization on a startup office workstation. Perform a clean install of Windows/Linux from ISO. Manually partition the disk (not guided/automatic partitioning). Optimize startup programs/services for faster boot. OS must boot without errors after installation.
ANS:

Task 4: Clean OS Installation

1. Prepare the installation media

- Download a Windows or Linux ISO from the official website.
- Create a bootable USB using Rufus or Balena Etcher.
- Back up important data before installation.

2. Perform a clean installation

- Boot the workstation from the USB drive.
- Select Install OS and choose the custom/manual installation option.
- Select the target disk and create partitions manually.

3. Manual disk partitioning example

For a 256 GB SSD:

Partition	Size	Purpose
EFI System Partition	512 MB	UEFI boot files
OS partition (`C:` or `/`)	180 GB	Operating system and applications
Data partition (`D:` or `/home`)	Remaining space	User files

Partition requirements vary by OS. Windows may create additional recovery partitions automatically; Linux may use a swap file instead of a dedicated swap partition.

4. Optimize startup

- Install the latest compatible drivers and OS updates.
- Disable unnecessary startup applications using Task Manager → Startup apps on Windows.
- On Linux, review startup applications and unnecessary services; do not disable essential system services.
- Keep security software enabled.

5. Verify installation

- Restart the workstation.
- Confirm that the OS boots without errors.
- Check Device Manager or system logs for hardware or driver errors.
- Confirm that storage, network, and startup applications work correctly.

Mini Project: Nexus Innovations — Diagnosing Random System Reboots Objective: At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module. Your project must: Identify the three hardware components most likely to cause random reboots under load.

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

1. Three Hardware Components Most Likely to Cause Random Reboots

1. CPU (Processor): Heavy rendering increases CPU temperature. Overheating can cause the system to shut down or restart to prevent damage.
2. GPU (Graphics Card): Rendering puts a high load on the GPU, which may overheat or become unstable.
3. PSU (Power Supply Unit): The PSU may fail to provide sufficient or stable power when the CPU and GPU consume more power under heavy workloads.

2. Why Systems Pass POST in the Morning but Fail in the Afternoon

POST (Power-On Self-Test) checks basic hardware functionality during startup. In the morning, the workstations are relatively cool and can pass POST successfully.

During afternoon rendering, CPU and GPU temperatures rise. Poor ventilation inside desk cabinets traps hot air, causing overheating. At the same time, heavy workloads increase power consumption, potentially exposing PSU limitations or faults. These conditions can cause random restarts.

3. Using a Stress-Test Tool

Use monitoring software such as HWMonitor or HWiNFO to observe CPU/GPU temperatures, clock speeds, and power readings. Then perform controlled CPU and GPU stress tests using suitable tools such as OCCT.

Procedure:

1. Record temperatures and power readings at idle.
2. Run a CPU stress test, followed by a GPU test; test both together if safe and appropriate.
3. Monitor temperatures, power stability, errors, and whether the system restarts.
4. Stop the test if temperatures exceed manufacturer limits or the system becomes unstable.

Interpretation: If temperatures rise excessively before a restart, overheating is likely. If restarts occur under combined load without excessive temperatures, investigate PSU capacity, power delivery, and cabling. Software readings alone cannot conclusively prove a PSU fault.

4. Three Specific Remediation Steps

1. Improve ventilation: Remove workstations from enclosed desk cabinets, ensure clear airflow, clean dust filters, and verify that fans and CPU/GPU coolers work correctly.
2. Inspect and maintain hardware: Check CPU/GPU temperatures, cooler mounting, thermal paste condition, fan operation, power connectors, and RAM stability. Replace faulty components as needed.
3. Test power delivery and verify stability: Check the PSU's capacity and condition, inspect cables and wall power, run controlled stress tests, and review system logs for thermal or hardware errors. Update BIOS and drivers where appropriate.

5. Is a PSU Upgrade Necessary?

A PSU upgrade may be necessary if the existing unit cannot supply enough power for the CPU and GPU under peak rendering loads, is faulty, or has unstable power delivery. Select a reputable PSU with sufficient wattage and suitable connectors for the hardware.

However, an upgrade may not be required if the PSU is healthy and adequately rated. Since the workstations are inside poorly ventilated cabinets, thermal management should be investigated first, while PSU condition and capacity are checked before purchasing replacements.

Conclusion

The most likely causes are CPU/GPU overheating and unstable power delivery. Improving airflow, inspecting cooling components, monitoring temperatures, and testing the PSU under controlled workloads should help identify the root cause and prevent frequent system restarts.

STEP 1 · BUILD WITH AI Use an AI tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 systems and 1 printer. The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address. The plan must allocate addresses for all 5 systems and 1 printer named in the scenario. Instruct the AI to also identify which addresses to reserve for future expansion. Document the exact prompt(s) used. STEP 2 · TEST & DEBUG (WITHOUT AI) Then, working without AI, verify the plan for errors — overlapping ranges, incorrect network/broadcast addresses, insufficient host capacity, or unreserved growth space — and correct them yourself.
ANS:

Nexus Innovations — IP Addressing and Subnetting Plan

Step 1: Build with AI

Exact prompt used:

Generate an IP addressing and subnetting plan for Nexus Innovations' startup office network containing 5 systems and 1 printer. Specify the network address, subnet mask, usable host range, and broadcast address. Assign a unique static IP address to each system and the printer. Reserve additional IP addresses for future expansion. Use a private IPv4 address range and explain the plan in simple language.

1. Network addressing plan

Network detail	Value
Network address (subnet)	`192.168.1.0/24`
Subnet mask	`255.255.255.0`
Usable host range	`192.168.1.1 – 192.168.1.254`
Broadcast address	`192.168.1.255`
Total usable host addresses	254

2. IP address allocation

Device name	Assigned IP address	Purpose
System 1	`192.168.1.1`	Office workstation
System 2	`192.168.1.2`	Office workstation
System 3	`192.168.1.3`	Office workstation

System 4	`192.168.1.4`	Office workstation
System 5	`192.168.1.5`	Office workstation
Printer 1	`192.168.1.6`	Shared network printer

All six devices use subnet mask `255.255.255.0`.

3. Addresses reserved for future expansion

- `192.168.1.7 – 192.168.1.20`: Reserve for additional computers and printers.
- `192.168.1.21 – 192.168.1.50`: Reserve for future office devices and servers.
- `192.168.1.51 – 192.168.1.254`: Keep available for additional devices or planned network services, subject to the network's IP allocation policy.

These are reservations for planning, not separate subnets. Ensure that reserved addresses are not assigned to other devices.

Step 2: Test and Debug (Without AI)

Use this checklist to verify the plan yourself.

Manual verification checklist

0 of 6

Check the network address

192.168.1.0 is the network address; do not assign it to a device.

Check the broadcast address

192.168.1.255 is the broadcast address; do not assign it to a device.

Check host capacity

A /24 subnet provides 254 usable addresses, enough for 6 devices and future growth.

Check for duplicate IPs

All six assigned device addresses are unique.

Check future reservations

The addresses .7 through .50 are reserved for planned expansion and must be excluded from other allocations.

Check for overlapping ranges

All devices and reservations belong to the same 192.168.1.0/24 subnet; there are no overlapping subnet definitions.

Correction result: The proposed plan passes the basic subnet, capacity, unique-address, and broadcast checks. No address correction is required.

Conclusion: Nexus Innovations can use the `192.168.1.0/24` network for its five systems and one printer, with additional addresses reserved for future expansion.